NETSAGE AI

PROJECT SUMMARY DOCUMENT

AI-Assisted Network Troubleshooting & Responsible Human Review System

PROJECT NAME: **NETSAGE AI**

COLLEGE - **Lakshmi Narain College Of Technology**

TEAM MEMBERS:

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Project Links

GitHub Repository: https://github.com/amansahu04444-cmd/NetSageAI

Live Demo: netsageai · main · streamlit_app.py

NetSage AI — AI-Assisted Network Troubleshooting Intelligence

AI-powered network diagnosis with deterministic rule checking, human-in-the-loop validation, and responsible AI governance.

2. Project Overview

NetSage AI is an AI-assisted network troubleshooting system designed to help network engineers, students, and administrators analyze Cisco/Packet Tracer network problems.

The system accepts structured network troubleshooting cases containing information such as:

- Network issue description
- Device configuration
- CLI output
- Connectivity symptoms
- Routing information
- VLAN information
- ACL configuration
- DHCP/DNS information
- OSPF configuration
- Interface status
- Other network evidence

The system analyzes this information and generates a structured diagnosis containing:

- Root cause
- Diagnosis status
- Confidence score
- OSI layer
- Evidence
- Rule-checker findings
- Recommended next command
- Fix steps
- Explanation

A key feature of NetSage AI is that AI recommendations are not automatically executed.

Every AI recommendation can go through a mandatory human review process, where a reviewer can:

1. Accept the AI recommendation
2. Modify the AI recommendation
3. Reject the AI recommendation

All review actions are stored as audit records.

This makes NetSage AI a human-in-the-loop decision-support system rather than an autonomous network configuration system.

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3. Problem Statement

Network troubleshooting generally requires engineers to manually inspect:

- Device configurations
- CLI outputs
- Routing tables
- VLAN configuration
- Interfaces
- ACL rules
- Protocol states
- Connectivity symptoms

For students and junior network engineers, identifying the actual root cause can be difficult and time-consuming.

Traditional troubleshooting also has several challenges:

- Large amounts of CLI output
- Multiple possible root causes
- Difficulty identifying relevant evidence
- Lack of structured diagnosis
- Human errors during troubleshooting
- Lack of consistent documentation
- Difficulty evaluating AI-generated troubleshooting recommendations

Therefore, there is a need for a system that can assist humans in diagnosing network issues while keeping the human engineer responsible for the final decision.

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4. Proposed Solution

NetSage AI provides a structured troubleshooting workflow.

Basic workflow

Network Case → Evidence Analysis → Rule Checking → AI Diagnosis → Human Review → Final Decision → Audit Record → Analytics

The system combines two important approaches:

4.1 Deterministic Rule Checking

Known networking conditions are checked using predefined rules.

Examples:

- Interface down
- Missing VLAN
- Missing route
- Incorrect ACL
- Routing configuration issue
- Trunk configuration issue

4.2 AI-Based Diagnosis

Gemini analyzes the case and provides a structured diagnosis.

The AI output is validated against the expected structured schema.

4.3 Human Review

The human reviewer evaluates the AI recommendation.

The reviewer can:

- Accept
- Modify
- Reject

4.4 Audit & Analytics

Every review is recorded and used to calculate responsible-AI metrics.

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5. Objectives

The main objectives of NetSage AI are:

1. Automate initial network troubleshooting analysis.
2. Reduce the time required to identify potential root causes.

3. Provide evidence-based network diagnosis.
4. Identify the relevant OSI layer.
5. Provide recommended verification commands.
6. Provide suggested corrective steps.
7. Combine deterministic rules with AI reasoning.
8. Maintain human control over final decisions.
9. Maintain an auditable review history.
10. Measure AI-human agreement.
11. Measure human corrections and rejections.
12. Provide responsible-AI analytics.
13. Prevent automatic execution of network commands.
14. Support offline/mock testing when Gemini is unavailable.

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6. Target Users

NetSage AI can be used by:

6.1 Network Engineering Students

Students can understand network troubleshooting cases and learn how evidence maps to root causes.

6.2 Junior Network Engineers

The system can provide an initial diagnosis and suggested verification steps.

6.3 Network Administrators

Administrators can use the system as a decision-support tool.

6.4 Trainers / Faculty

Faculty can use structured network cases for teaching and evaluation.

6.5 AI/Network Research Teams

The human-review and audit system can be used to evaluate AI troubleshooting reliability.

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7. Key Features

7.1 Network Case Dataset

NetSage AI uses a structured network troubleshooting dataset.

The current project dashboard reports:

- 30 dataset cases
- 20 issue categories

The cases cover multiple networking areas.

Examples include:

- ACL
- Addressing
- CDP
- DHCP
- DNS
- HSRP
- IPv6
- Inter-VLAN Routing
- NAT
- OSPF
- Port Security
- Security/DAI
- Static Routing
- Subnetting
- Switching
- VLAN
- VLAN Trunking
- VTP
- Wireless
- Wireless/ACL

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8. AI Diagnosis

The AI diagnosis engine generates a structured diagnosis.

The diagnosis contains:

Case ID

Identifies the network troubleshooting case.

Root Cause

The suspected or confirmed underlying cause.

Diagnosis Status

The diagnosis can indicate states such as:

- CONFIRMED
- LIKELY

Confidence

A numerical confidence score between:

0.0 and 1.0

OSI Layer

The affected networking layer.

Examples:

- Layer 2
- Layer 3
- Layer 4

Evidence

The specific network evidence supporting the diagnosis.

Rule Checker Findings

Findings generated by deterministic troubleshooting rules.

Next Command

A command that a human engineer can execute for verification.

Fix Steps

Suggested steps for resolving the issue.

Explanation

A human-readable explanation of the diagnosis.

Mock Flag

The diagnosis includes:

is_mock

This indicates whether the diagnosis was generated through offline/mock mode.

9. Gemini AI Integration

NetSage AI integrates Google's Gemini model for AI-assisted diagnosis.

The project uses Gemini to analyze network troubleshooting evidence and generate structured diagnostic recommendations.

The Gemini integration is designed to:

1. Receive network troubleshooting information.
2. Analyze the evidence.
3. Identify a potential root cause.
4. Determine diagnosis status.
5. Estimate confidence.
6. Identify the relevant OSI layer.
7. Provide evidence.
8. Suggest verification commands.
9. Provide fix steps.
10. Generate an explanation.

The current working integration uses the configured Gemini model.

Important: The Gemini integration is not modified as part of the human-review data cleanup process.

10. Structured AI Output

The AI output follows a structured DiagnosisResult schema.

The schema contains:

case_id root_cause

diagnosis_status

confidence

osi_layer evidence

rule_checker_findi

ngs

next_command

fix_steps

explanation is_mock

This prevents the system from relying on unstructured AI text alone.

Structured output makes it easier to:

- Validate AI responses
- Store diagnoses
- Display results
- Compare AI and human decisions
- Audit results
- Generate analytics

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11. Rule Checker

NetSage AI includes a deterministic rule-checking component.

The rule checker provides additional evidence independent of the AI reasoning.

Example: Missing

VLAN Rule:

missing_vlan

Status:

ERROR

Message:

Required VLAN 20 is missing from trunk allowed list

Another example:

Missing Route

Rule:

missing_route

Status:

ERROR

Message:

Passive interface enabled on active OSPF link blocking routes

This creates an additional verification layer around AI diagnosis.

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12. Human-in-the-Loop Review

One of the most important features of NetSage AI is mandatory human review.

The system does not treat the AI recommendation as the final answer automatically.

The reviewer evaluates the recommendation.

There are three primary decisions.

12.1 Accept

The reviewer confirms that the AI recommendation is correct.

Example:

Status: ACCEPTED

Reviewer: Aman

Reason: Looks accurate.

12.2 Modify

The reviewer believes the diagnosis is generally useful but needs improvement.

The reviewer can modify the diagnosis.

Example:

Status: MODIFIED

Reviewer: Aman (Lead Engineer)

Reason:

Refined OSPF root cause and added explicit neighbor and

route verification steps.

12.3 Reject

The reviewer determines that the AI diagnosis should not be accepted.

Example:

Status: REJECTED

Reason:

The AI diagnosis is incomplete; the ACL evidence requires human verification before accepting the proposed root cause.

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13. Review Record

Each human review is stored as an audit record. The review structure contains information such as:

review_id

case_id status

ai_diagnosis

human_diagnosis

reviewer reason

created_at

reviewed_at

This provides traceability for every review decision.

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14. Review Statuses

The system supports:

ACCEPTED

AI recommendation accepted as presented.

MODIFIED

AI recommendation changed by human reviewer.

REJECTED

AI recommendation rejected.

PENDING

Human review has not yet been completed.

15. Audit Logging

The project maintains review records in:

review/reviews.json

The audit system allows the project to track:

- Who reviewed the diagnosis
- What decision was made
- Why the decision was made
- Which case was reviewed
- When the review was created
- When it was completed
- Whether the original AI diagnosis was modified

This is important for responsible AI governance.

16. Responsible AI Design

NetSage AI follows a human-controlled decision-support approach.

The system explicitly states:

NetSage AI operates strictly as a passive decision-support tool.

AI recommendations do not automatically execute Cisco IOS commands.

The final network configuration decision remains with a human engineer.

17. Safety Restriction

The system does not automatically:

- Execute Cisco IOS commands
- Modify routers
- Modify switches
- Change VLAN configuration
- Change routing configuration
- Modify ACLs
- Shut/no-shut interfaces
- Alter network devices

Instead, it provides recommendations that require human action.

For example, the AI may recommend:

configure terminal interface

GigabitEthernet0/0.10 no

shutdown

But the system does not execute these commands automatically.

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18. Offline Mock Mode

NetSage AI also supports offline/mock mode.

Mock mode is useful when:

- Gemini API is unavailable
- API key is not configured
- Development is being performed
- Automated tests are running
- Demonstration data is required
- Internet/API access is unavailable

Mock diagnoses are identified through: is_mock:

true

This makes it possible to distinguish demo/test results from real Gemini results.

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19. Streamlit Web Interface

NetSage AI provides a Streamlit-based web interface.

The application can be launched using:

python -m streamlit run streamlit_app.py This

command is preferable on systems where:

streamlit run streamlit_app.py is

not recognized.

The interface runs locally in the browser.

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20. AI Configuration Panel

The Streamlit sidebar provides AI configuration information.

It displays whether Gemini is configured.

It also provides an option for:

Use Offline Mock Mode

This allows users to switch between:

Mock Mode

Offline deterministic/demo diagnosis.

Gemini Mode

Real Gemini-based diagnosis.

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21. Reviewer Identity

The application provides a reviewer identity field.

This allows review records to identify who performed the human review.

Example:

Aman (Lead Engineer)

This improves auditability.

22. Executive Dashboard

NetSage AI includes an executive analytics dashboard.

The dashboard provides high-level project metrics.

Current cleaned dataset state showed:

Total Dataset Cases : 30

Categories Covered : 20

Total Review Records : 2

Completed Human Reviews : 1

Pending Reviews : 1

AI Accepted Count : 1

AI Modified Count : 0

AI Rejected Count : 0

AI-Human Agreement Rate : 100.0%

Human Correction Rate : 0.0%

Human Rejection Rate : 0.0%

Corrected AI Responses : 0

These values are based on the currently retained real Gemini review records.

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23. KPI Metrics

The dashboard tracks:

Total Dataset Cases

Number of network troubleshooting cases.

Categories Covered

Number of different network issue categories.

Total Review Records

Total human-review records.

Completed Human Reviews

Number of reviews that are no longer pending.

Pending Reviews

Number of reviews awaiting human action.

AI Accepted Count

Number of reviews where AI diagnosis was accepted.

AI Modified Count

Number of reviews where AI diagnosis was modified.

AI Rejected Count

Number of reviews where AI diagnosis was rejected.

AI-Human Agreement Rate

Percentage of completed reviews where the human accepted the AI recommendation.

Human Correction Rate

Percentage of completed reviews where the human modified the AI recommendation.

Human Rejection Rate

Percentage of completed reviews where the human rejected the AI recommendation.

Corrected AI Responses

Total number of AI responses that required human correction or rejection, according to the dashboard's metric definition.

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24. Responsible AI Analytics

The dashboard also provides responsible-AI metrics.

These include:

AI-Human Agreement Rate

Measures how frequently humans accept AI recommendations.

Human Correction Rate

Measures how frequently humans modify AI recommendations.

Human Rejection Rate

Measures how frequently humans reject AI recommendations.

Total Corrected AI Responses

Measures the number of AI outputs that required human intervention.

These metrics help evaluate the reliability of AI-generated network diagnoses.

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25. Data Origin Tracking

The system distinguishes between:

Real Gemini Responses

Diagnoses generated using the real Gemini integration.

Mock/Demo Responses

Diagnoses generated using offline/mock mode or automated testing.

The is_mock field is used to identify the origin.

This is particularly important because demo/test data should not be mixed with real production-style AI evaluation results.

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26. Review Data Cleanup

During development, the project generated multiple automated test/demo review records.

The cleanup process identified:

Total Records Before Cleanup : 168

Mock/Demo Records Found : 166

Repeated NET-001 Mock : 158

Other Case Mocks : 8

Real Gemini Records : 2

The cleanup preserved the real Gemini records and removed only clearly identified mock/demo records.

The retained records were:

REV-B37599 | NET-001 | ACCEPTED

REV-72BC75 | NET-001 | PENDING

This ensures that final dashboard metrics are not artificially influenced by previous automated tests.

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27. Repeated NET-001 Test Data

During development, NET-001 was repeatedly used for automated testing.

The cleanup identified:

158 repeated NET-001 mock/test records

These records were identified as mock/demo records using the ai_diagnosis.is_mock field.

They were therefore excluded from the cleaned review dataset.

This prevents automated testing activity from being incorrectly interpreted as real human-AI evaluation.

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28. Cleaned Review Dataset

After cleanup:

Total Review Records: 2

Completed Human Reviews: 1

Pending Reviews: 1

This provides a much cleaner baseline for future real Gemini evaluation.

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29. Example Real Gemini Diagnosis

A real diagnosis can identify a network issue such as:

Root Cause:

Router sub-interface GigabitEthernet0/0.10 is administratively down, disabling the default gateway for VLAN 10 and breaking inter-VLAN routing.

The system can identify:

Diagnosis Status:

CONFIRMED

Confidence:

1.00

OSI Layer:

Layer 2/3

It can provide evidence such as:

GigabitEthernet0/0.10 is administratively down line protocol is down

And recommend verification/fix steps.

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30. Example Recommended Fix

For an interface-down problem, the system may recommend:

1. Enter global configuration mode.
2. Enter the affected sub-interface.
3. Enable the interface using no shutdown.
4. Verify the main interface if necessary.
5. Verify interface status using show ip interface brief.

Again, these commands are recommendations only.

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31. Example NET-008 Case

One reviewed case involved a VLAN trunk issue.

The diagnosis identified:

Required VLAN 20 is missing from trunk allowed list.

Evidence included: switchport trunk

allowed vlan 10 30 40 where VLAN

20 was missing.

The human reviewer modified the diagnosis to clarify the root cause and add verification.

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32. Example NET-005 Case

Another case involved an ACL-related issue.

The AI diagnosis identified a potential problem related to:

Web server port 80 unreachable from Sales subnet and

identified an ACL entry denying TCP traffic to port 80.

The human reviewer rejected the diagnosis because the ACL evidence required further verification.

This demonstrates why the human-review layer is important**.**

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33. Example NET-012 Case

Another case involved OSPF.

The diagnosis identified:

Passive interface enabled on active OSPF link

The human reviewer modified the diagnosis and added explicit:

- OSPF neighbor verification
- Route verification

This demonstrates how human expertise can improve an AI-generated troubleshooting recommendation.

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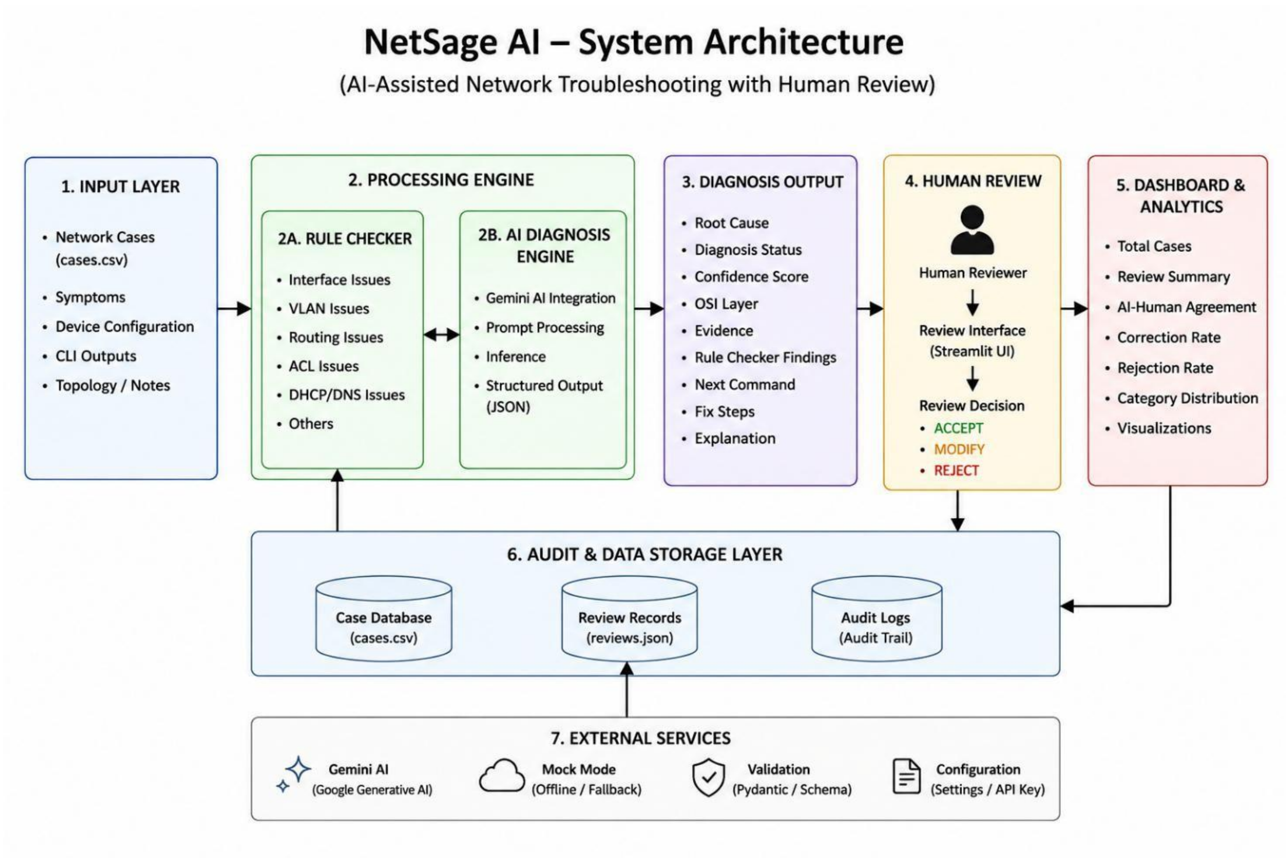
34. System Architecture

The project can be represented as:



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35. Project Folder Structure

Current project structure:

```
NetSage-AI/
│
├── ai/
│   ├── diagnosis_engine.py
│   └── structured_output.py
│
├── checker/
│   └── ...
│
├── dashboard/
```

```
│   └── ...
│
├── data/
│   └── cases.csv
│
├── prompts/
│   └── ...
│
├── review/
│   ├── reviews.json
│   ├── review_service.py
│   └── audit_logger.py
│
├── scripts/
│   └── clean_demo_reviews.py
│
├── app.py
├── streamlit_app.py
├── README.md
├── requirements.txt
└── .gitignore
```

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36. Technologies Used

- Programming Language
- Python
- AI
- Google Gemini
- AI Integration
- Google GenAI API
- Frontend / Dashboard
- Streamlit
- Data
- CSV / JSON
- Validation
- Pydantic
- Development
- VS Code / Antigravity IDE
- Version Control

- Git / GitHub

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37. Software Components

The project contains multiple logical components.

AI Module

Responsible for:

- Gemini communication
- Diagnosis generation
- Mock diagnosis
- Structured output Checker Module

Responsible for:

- Deterministic network rule checking
- Identifying known configuration issues

Review Module

Responsible for:

- Creating review records
- Accepting reviews
- Modifying reviews
- Rejecting reviews
- Listing reviews
- Calculating review metrics Dashboard Module Responsible for:
- KPI analytics
- Review statistics
- Responsible-AI analytics
- Data visualization

Data Module

Responsible for:

- Network troubleshooting cases

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38. CLI Operations

The project supports several CLI operations.

Validate Dataset python app.py --validate

Check Rules python app.py --check-rules

Diagnose Case python app.py --diagnose NET-001 **Diagnose in Mock Mode** python app.py --diagnose NET-001 --mock **Create Review** python app.py --review NET-001

List Reviews python app.py --list-reviews

Dashboard Metrics python app.py --dashboard-metrics

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39. Running the Web Application

The Streamlit application can be started with:

python -m streamlit run streamlit_app.py

Then open the local Streamlit URL shown in the terminal, normally: http://localhost:8501

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40. Testing Workflow

A recommended testing workflow is:

Start Application

↓

Select Network Case

↓

Run Diagnosis

↓

Review AI Recommendation

↓

Check Evidence

↓

Check Rule Findings

↓

Human Decision

↓

Accept / Modify / Reject

↓

Audit Record Created

↓

Dashboard Metrics Updated

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41. Human Review Example

AI Recommendation

Root Cause:

Interface GigabitEthernet0/0.10 is administratively down.

Human Reviewer

Reviewer:

Aman (Lead Engineer)

Decision

ACCEPTED

Reason

Confirmed from CLI evidence.

The decision becomes part of the audit history.

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42. Advantages

NetSage AI provides several advantages.

Faster Troubleshooting

AI helps identify potential causes quickly.

Evidence-Based Diagnosis

The system presents supporting evidence.

Rule + AI Combination

Deterministic checks provide an additional layer of validation.

Human Control

Humans make the final decision.

Auditability

Review decisions are stored.

Responsible AI

The system measures AI-human agreement and corrections.

Offline Testing

Mock mode allows development without API access.

Structured Output

AI responses follow a predictable schema.

Educational Value

Students can learn troubleshooting from structured cases.

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43. Limitations

The project also has limitations.

1. Diagnosis quality depends on the quality of network evidence.
2. AI predictions may still be incorrect.
3. Human review is required for reliable final decisions.
4. Gemini API availability may affect real-AI operation.
5. Mock mode does not represent real Gemini reasoning.
6. The current dataset contains a limited number of cases.
7. The system does not directly configure network devices.
8. Recommended commands require manual verification and execution.
9. Real-world network environments may contain conditions not represented in the dataset.

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44. Security & Safety

NetSage AI follows a passive troubleshooting model.

The system does not automatically execute network commands.

This reduces the risk of:

- Incorrect configuration

- Accidental device shutdown
- Routing disruption
- VLAN changes
- ACL modification
- Network outages

The human engineer remains responsible for validating recommendations.

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45. Future Scope

Possible future improvements include:

45.1 Larger Dataset

Add more real-world troubleshooting cases.

45.2 Multi-Device Analysis

Analyze multiple routers and switches together.

45.3 Advanced Network Topology

Visualize network topology and affected paths.

45.4 Historical Learning

Use reviewed cases to improve future troubleshooting assistance.

45.5 Advanced Evaluation

Compare multiple AI models.

45.6 Confidence Calibration

Improve confidence-score reliability.

45.7 Role-Based Review Introduce roles such as:

- Network Engineer
- Senior Engineer
- Auditor
- Administrator

45.8 Advanced Audit System

Add immutable audit trails and richer review history.

45.9 Enterprise Deployment

Deploy the system as an internal network troubleshooting platform.

46. Project Impact

NetSage AI demonstrates how generative AI can be applied to network troubleshooting while maintaining human oversight.

The project combines:

Networking + Artificial Intelligence + Rule-Based Reasoning + Human-in-the-Loop + Responsible AI + Analytics

This makes the system more suitable for practical decision support than a simple chatbot that only generates troubleshooting text.

47. Conclusion

NetSage AI is an AI-assisted network troubleshooting platform that combines Gemini-based reasoning with deterministic network rule checking and mandatory human review.

The system transforms raw network troubleshooting evidence into structured diagnostic recommendations.

The human reviewer remains responsible for the final decision through:

Accept

Modify

Reject

All decisions are stored in an audit system and analyzed through responsible-AI metrics.

The project therefore demonstrates a practical approach to using generative AI in networking while maintaining:

- Human oversight
- Safety
- Explainability
- Evidence-based reasoning
- Auditability
- Responsible AI governance